Elliptic curves

Pushpendra Pal

June 2025

Definitions

- **Algebraic Curve:**
  A curve $\mathbb{C}(x_1, x_2, \ldots, x_n) = 0$
  $x_i \in \mathbb{F}$ (finite or infinite)
  For e.g., circle, ellipse, etc.

- **Algebraically Closed Field:**
  A field $\mathbb{F}$ is algebraically closed if it contains the root of every non-constant polynomial in $\mathbb{F}$[x], where $\mathbb{F}$[x] is the ring of polynomials in varaible x with $\mathbb{F}$ coefficients.

- **Singular Curve**:
  A curve $\mathbb{C}(x_1, x_2, \ldots, x_n)$ is singular if $\exists$ a point 'P' in its domains where

$$\mathbb{C}(p) = \mathbb{C}'_{\mathbb{1}}(p) = \mathbb{C}'(p) = \ldots = \mathbb{C}'(p) = 0$$

Different types

General Weierstrass Form

- $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$
- Works in any characteristic, including 2,3
- Encompasses all elliptic curves, as long the $\Delta \neq 0$.

Short Weierstrass Form

- $y^2 = x^3 + ax + b$
- Valid when characteristic $\neq$ 2,3.
- used in most of cryptography over prime fields $\mathbb{F}_p$. Bitcoin, ethereum (secp256k1) NIST curves.

Binary curves

- $y^2 + xy = x^3 + ax^2 + b$
- Works for characteristic 2, e.g. $\mathbb{F}_{2^m}$
- Used in constraint environments (Short codes, RFID).

Montgomery curves

- $By^2 = x^3 + Ax^2 + x$
- Used in x25519 (Diffie hellman KEX)
- Efficient for scalar multiplcation.

Edwards curves

- $y^2 + x^2 = 1 + dx^2y^2$
- Used in ED25519
- Offers complete, fast addition formulae

Twisted edwards curves

- $ax^2 + y^2 = 1 + dx^2y^2$
- Generalization of edwards
- Fast and complete group laws

General Weierstrass Form

- $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$
- Works in any characteristic, including 2,3
- Encompasses all elliptic curves, as long the $\Delta \neq 0$.

Properties

- **Composition law**

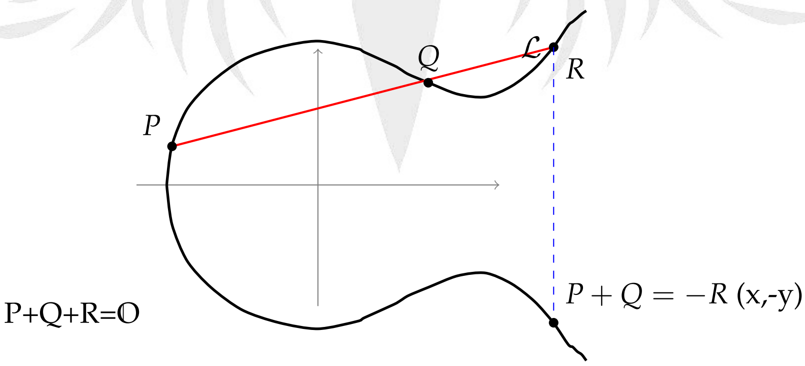

Figure 1: A line $\mathcal{L}$ (positive slope) intersecting the elliptic curve at P, Q, R. The vertical dashed line through R meets the lower branch at $-R$.

- **Identity**: P + $\mathbb{O}$ = P
- **Commutativity**: P + Q = Q + P
- **Inverse**: If P ϵ $\mathbb{E}$, then $\exists$ (-P) ϵ $\mathbb{E}$,

$$P + (-P) = \mathbb{O}$$

- **Associativity** (P+Q) + R = P + (Q + R)
- **Composition**: If a straight line intersects $\mathbb{E}$ at three points namely, P Q and R, then

$$P + Q + R = \mathbb{O}$$
$$P + Q = -R$$

where (-R is basically the reflection of point R across the x-axis)

Discriminant

Theorem 1: $\Delta = 0 \equiv$ Elliptic curve $\mathbb{E}$ is singular
Proof: $\mathbb{E}$ is singular $\implies$ Δ = 0

- Let $\exists$P $\epsilon(\alpha, \beta)$ on $\mathbb{C}$ such that
  $\mathbb{E}$(P) = 0, $\mathbb{E}_x(P) = 0$, $\mathbb{E}_y(P) = 0$
- $\mathbb{E}(P) = -y^2 + (x^3 + Ax + B) = 0$
  $\mathbb{E}_x(P) = 3x^2 + A = 0.$
  $\mathbb{E}_y(P) = -2y = 0$
  $\mathbb{E}_y(P)$ gives y=0.
- Put y=0 in $\mathbb{E}(P)$, then $\mathbb{E}(P) = (x^3 + Ax + B)$
  So P = (α,0) is a root of both $\mathbb{E}(P)$ and $\mathbb{E}_x(P)$, hence a double root

$$\begin{aligned} x^3 + Ax + B &= (x-\alpha)^2(x-\gamma) \\ &= x^3 + (-2\alpha - \gamma)x^2 + (\alpha^2 + 2\alpha\gamma)x - \alpha^2\gamma \end{aligned}$$

$(2\alpha + \gamma)x^2 = 0 \implies \gamma = -2\alpha \ldots$ by comparing with coeff of x^2 in $x^3 + Ax + B$
Let's substitute γ in $x^3 + (\alpha^2 + 2\alpha\gamma)x - \alpha^2\gamma$, we get

$$\begin{aligned} A &= (\alpha^2 + 2\alpha(-2\alpha)) && = -3\alpha^2 \\ B &= -\alpha^2(-2\alpha) && = 2\alpha^3 \end{aligned}$$

- Put this in equation of $\Delta = -16(4A^3 + 27B^2)$
  $\Delta = -16(4(-3\alpha^2)^3 + 27(2\alpha^3)^2)$
  $\Delta = -16(-108\alpha^6 + 108\alpha^6)$
  $\Delta = 0$

Proof: Δ = 0 $\implies$ $\mathbb{E}$ is singular
Contrapositive: $\mathbb{E}$ is not singular $\implies$ $\Delta \neq 0$

- Let's consider the homogenous form of elliptic curve:

$$Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3$$

- $\mathbb{O}$ = [0,1,0] is the poitn at infinity, then
- $E_z(\mathbb{O}) = Y^2 = 1 \neq = 0$
- Hence $\mathbb{E}$ is non singular at $\mathbb{O}$.
- Here, it is slightly confusing. But what I understand is if we have a non singular curve in homogenous form, then $\Delta \neq 0$

Defining Addition with P(x_1, y_1) and Q(x_2, y_2)

Case 1: $x_1 \neq x_2$, $y_1 \neq y_2$

- Take line $\mathbb{L}$: y = λx + v, passing thru P, Q
- $\lambda = \frac{y_2 - y_1}{x_2 - x_1}$
- $y^2 = (\lambda x + v)^2 = x^3$ + Ax + B
- x^3 - $\lambda^2 x^2$ + (A-2 λv)x + B-v^2
- So, $\lambda^2 = x_1 + x_2 + x_3$ (Sum of roots)
  $x_3 = \lambda^2$ - x_1 - x_2
- $y_t = \lambda(x_3$ - $x_1)$ + y_1, but we want the negation of y_3, so
  that means the point should be a reflection of R = (P+Q) across the X-axis.
  $y_3 = \lambda(x_1$ - $x_3)$ - y_1,

Case 2: $x_1 = x_2$, $y_1 = y_2$, **tangent line**

- $\frac{d(y^2)}{dx} = \frac{d(x^3+Ax+B)}{dx}$
- $\lambda = \frac{3x^2+A}{2y}$